Question 1 : Define Data Transformation in ETL and explain why it is important

Data Transformation is the process of converting raw data extracted from various source systems into a clean, consistent, and structured format suitable for analysis and loading into a target system (such as a data warehouse).

Data Transformation Important -

- **Improves Data Quality -** Ensures the data is accurate, complete, and consistent.
- **Enables Better Decision-Making -** Clean and structured data allows organizations to generate reliable reports and insights.
- **Ensures Consistency Across Systems -** Standardizes data from different sources into a unified format.
- **Supports Business Rules -** Applies organizational logic (e.g., calculating profit, categorizing customers).
- **Enhances Performance -** Well-structured data improves query performance in data warehouses.
- **Ensures Compliance and Governance -** Helps maintain regulatory standards by validating and securing data.

Question 2 : List any four common activities involved in Data Cleaning.

Four Common Activities Involved in Data Cleaning

- **Removing Duplicate Records -** Identifying and eliminating repeated entries to avoid inaccurate analysis and inflated results.
- **Handling Missing Values -** Filling in missing data (using mean/median/default values) or removing incomplete records when necessary.
- **Correcting Inconsistent Data -** Standardizing formats such as dates (e.g., DD/MM/YYYY vs MM/DD/YYYY), text capitalization, or measurement units.
- **Fixing Errors and Invalid Data -** Detecting and correcting incorrect values (e.g., negative age, invalid email formats, out-of-range numbers).

Question 3 : What is the difference between Normalization and Standardization?

Feature	Normalization	Standardization
Definition	Rescales data to a fixed range, usually between 0 and 1	Rescales data so it has a mean of 0 and standard deviation of 1
Formula	(X − Min) / (Max − Min)	(X − Mean) / Standard Deviation
Range of Output	Typically between 0 and 1	No fixed range (can be positive or negative)
Effect on Distribution	Changes the scale but not necessarily the distribution shape	Centers the data around zero and adjusts spread
Use Case	When data does not follow a normal distribution or when bounded range is required	When data follows (or is assumed to follow) a normal distribution

Question 4 : A dataset has missing values in the "Age" column. Suggest two techniques to handle this and explain when they should be used.

Here are two common techniques to handle missing values in the *Age* column:

1. Mean/Median Imputation
Replace missing age values with the mean (average) or median of the existing age values.

When to Use:

- Mean → When the age data is normally distributed (no extreme outliers).
- Median → When the data contains outliers or is skewed (median is more robust).
- When only a small percentage of values are missing.
- When maintaining dataset size is important.

Example:
If the average age is 30, replace missing ages with 30.

2. Deletion (Removing Rows with Missing Values)
Remove records (rows) where the Age value is missing.

When to Use:

- When only a very small number of records have missing ages.
- When missing data is random.
- When dataset size is large enough that removing rows won't affect analysis significantly.

Question 5 : Convert the following inconsistent "Gender" entries into a standardized format ("Male", "Female"): ["M", "male", "F", "Female", "MALE", "f"]

Given values:
["M", "male", "F", "Female", "MALE", "f"]

Standardized Format:

- Male → "M", "male", "MALE"
- Female → "F", "Female", "f"

Final Standardized Output:

["Male", "Male", "Female", "Female", "Male", "Female"]

Question 6: What is One-Hot Encoding? Give an example with the categories: "Red, Blue, Green".

One-Hot Encoding is a technique used to convert categorical data into numerical format by creating separate binary (0 or 1) columns for each category.

Each category gets its own column:

- 1 → Category present
- 0 → Category not present

Color	**Red**	**Blue**	**Green**
Red	1	0	0
Blue	0	1	0
Green	0	0	1

Question 7 : Explain the difference between Data Integration and Data Mapping in ETL.

Feature	Data Integration	Data Mapping
Definition	Combining data from multiple sources into a unified view	Defining how fields from source systems correspond to fields in the target system
Purpose	Create a consolidated dataset	Establish relationships between source and target data fields
Scope	Broader process	A step within integration
Example	Merging sales data from two databases	Mapping “cust_id” (source) to “customer_id” (target)

Question 8 : Explain why Z-score Standardization is preferred over Min-Max Scaling when outliers exist.

Min-Max Scaling rescales data between 0 and 1 using the minimum and maximum values. If outliers are present, they significantly affect the minimum and maximum, compressing most data into a very small range.

Z-score Standardization uses mean and standard deviation:

$$Z = (X - \mu)/\sigma$$

Why it is better with outliers:

1. It does not rely solely on minimum and maximum values.
2. Outliers do not compress the entire dataset into a narrow range.
3. It preserves the distribution shape better.